Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/29/2015
PI Name:	Conkin, Johnny Ph.D.		
Project Title:	Data Mining and Development of Spaceflight DCS Treatment Model		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	johnny.conkin-1@nasa.gov	**Fax:**	FY 281-483-3058
PI Organization Type:	NON-PROFIT	**Phone:**	281-244-1121
Organization Name:	Universities Space Research Association		
PI Address 1:	Biomedical Research and Environmental Sciences		
PI Address 2:	2101 NASA Parkway, SK3		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3696	**Congressional District:**	22
Comments:	PI previous affiliation (unknown when changed to USRA/JSC): Baylor College of Medicine.		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/29/2013	**End Date:**	03/31/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Jan 2014 report: Addition of Joseph P. Dervay, MD, as a co-investigator.		
COI Name (Institution):	Abercromby, Andrew Ph.D. (Wyle Science, Technology & Engineering) Feiveson, Alan Ph.D. (NASA Johnson Space Center) Gernhardt, Michael Ph.D. (NASA Johnson Space Center) Norcross, Jason M.S. (Wyle Science, Technology & Engineering) Ploutz-Snyder, Robert Ph.D. (Universities Space Research Association) Wessel, James M.S. (Wyle Science, Technology & Engineering) Dervay, Joseph M.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Hypobaric Decompression Sickness (DCS) Treatment Model links a decrease in computed bubble volume from increased pressure (delta P), increased oxygen (O2) partial pressure, and passage of time during treatment to the probability of symptom resolution [P(symptom resolution)]. The decrease in volume is realized in 2 stages: a) during compression of the offending gas phase (Boyle's Law) and b) during subsequent dissolution of the gas phase by the O2 window. The P(symptom resolution) during repressurization was modeled as a log logistic function of pressure difference (delta P as psid) and 2 other explanatory variables while accounting for multiple symptoms within subjects. We used data on 154 symptoms from 119 subjects with DCS in 47 different denitrogenation protocols (969 exposures) to fit the model. The fitted model was P(symptom resolution) = 1 / (1+exp(-(ln(deltaP) – 1.682 + 1.089×AMB - 0.00395×Ts) / 0.633)), where AMB = 1 if ambulation took place during part of the altitude exposure, otherwise AMB = 0; and where Ts is the elapsed time in min from start of the altitude exposure to recognition of a DCS symptom. To apply this model in future scenarios, values of delta P as inputs to the model would be calculated from the Tissue Bubble Dynamics Model based on the effective treatment pressure: P2 – P1 (delta P, psid) = P1×V1/V2 – P1, where V1 is the computed volume of a spherical bubble in a unit volume of tissue at low pressure P1 and V2 is computed volume after a change to a higher pressure P2. If 100% ground level O2 (GLO) was breathed in place of air, then V2 continues to decrease through time at P2 at a faster rate. The computed delta P is the effective treatment pressure at any point in time. Simulation of a "pain-only" symptom at 203 min into an ambulatory extravehicular activity (EVA) at 4.3 psia on Mars resulted in a P(symptom resolution) of 0.47 (0.36 to 0.58 95% confidence intervals) on immediate return to 8.2 psia in the Multi-Mission Space Exploration Vehicle. The P(symptom resolution) increased to near certainty (0.99) after 2 hrs of GLO at 8.2 psia or with less certainty on immediate pressurization to 14.7 psia [0.83 (0.75 – 0.89)]. Given the low probability of DCS during EVA and the prompt treatment of a symptom with guidance from the model it is likely that the symptom and gas phase will resolve with minimum resources and minimal impact on astronaut health, safety, and productivity. Executive Summary Decompression sickness (DCS) is an occupational hazard as long as extravehicular activity (EVA) is performed at suit pressure less than tissue inert gas tension. Efforts to eliminate DCS in astronauts through engineering control of the habitat atmosphere to minimize atmospheric nitrogen partial pressure will reduce the probability of DCS [P(DCS)] and severity of symptoms related to evolved gas. Prudent planning requires that DCS treatment resources be provided for Exploration Class missions where EVAs will be numerous, energetic, and a return to definitive medical care is not possible. Two statistical regression models are described that enable Mission Managers to quantify the P(DCS) associated with future EVAs and to quantify the probability of symptom resolution [P(symptom resolution)] given options for treatment. A biophysical Tissue Bubble Dynamics Model that models bubble growth and dissolution is linked to the 2 probability regression models. The regression models are quantitative descriptions of data collected during NASA-funded research on DCS from 1983 to 2014, about 1000 altitude exposures that evaluated different denitrogenation protocols prior to simulated EVAs in hypobaric chambers. A user of this "system" computes the P(DCS) and P(symptom resolution) given information about the denitrogenation (prebreathe) protocol, the suit pressure, the time to onset of a symptom during EVA, body mass index and age of the astronaut, and whether or not ambulation is present during the EVA. The ambulation status is an important variable in the regressions, distinguishing between ambulation during EVA on a planet surface or nonambulation during EVA in space. Effective treatment to achieve a high P(symptom resolution) is when treatment starts shortly after a symptom is recognized. Effective treatment is through the application of pressure during repressurization to the habitat pressure, an increase in habitat pressure, additional pressurization from the suit above habitat pressure, and the use of 100% oxygen (O2) breathing to accelerate the natural process of bubble dissolution. Combinations of both treatment pressure and 100% O2 breathing through time to achieve greater than or equal to 0.75 P(symptom resolution) at the lower 95% confidence interval is proposed as a requirement to establish treatment resources that assures a successful treatment outcome for Exploration Class EVAs. Adjunctive pharmaceutical and supportive critical care therapy are also necessary, even given a low probability of serious DCS. Options for DCS treatment after Exploration Class EVAs must be based on evidence. An evidence-based approach ultimately matches the most effective treatment to the anticipated risks. Providing medical treatment at remote locations is both costly and impacts the success of the mission, so it must be effective. The "system" described in this communication meets the requirement of an evidence-based approach. Specific Aims: 1) Consolidate data on factors (interventions) that provide for effective hypobaric DCS treatment from the NASA Hypobaric Decompression Sickness Database, 2) assemble these data into an NASA Hypobaric DCS Treatment Database, 3) create a statistical model on the probability of effective treatment given inputs on factors that influence effective treatment, 4) consolidate data on interventions that provide for effective hypobaric DCS treatment from published literature and from the Wright-Patterson Air Force Research Laboratory Decompression Sickness Research Database into the NASA Hypobaric DCS Treatment Database, 5) evaluate the initial statistical model (NASA data only) against data from literature and the Air Force database, 6) optimize model using all available data, 7) evaluate the use of an available bubble growth index (BGI) model as a DCS treatment model and compare to statistical DCS treatment model, 8) deliver to NASA a draft quantitative model(s) for astronaut DCS treatment, 9) document the relevant data, literature reports, and treatment model(s), and finally 10) communicate our findings to a wider audience in the event there are applications outside of NASA.
Rationale for HRP Directed Research:	The proposed task is considered highly constrained research using experts within NASA, who have awareness of the expected in-flight resources and technical background and experience to operate the bubble growth index model.
Research Impact/Earth Benefits:	The NASA Hypobaric DCS Treatment Model is specific to NASA EVA applications, but the same model may be useful in other aerospace applications where hypobaric DCS is an occupational risk. For example, applications in military aviation where standard-of-care hyperbaric treatment may not be available and the model could be used to develop alternative treatment strategies.

Task Progress:	The task was completed on time and on budget. A NASA Technical report and a report published in the June issue of Aerospace Medicine and Human Performance complete the deliverables. See Bibliography.
Bibliography Type:	Description: (Last Updated: 03/14/2017)
Articles in Peer-reviewed Journals	Conkin J, Abercromby AFJ, Dervay JP, Feiveson AH, Gernhardt ML, Norcross JR, Ploutz-Snyder R, Wessel JH 3rd. "Hypobaric decompression sickness treatment model." Aerospace Medicine and Human Performance. 2015 Jun;86(6):508-17. http://dx.doi.org/10.3357/AMHP.4178.2015 , Jun-2015
Articles in Peer-reviewed Journals	Conkin J. "Equivalent air altitude and the alveolar gas equation." Aerosp Med Hum Perform. 2016 Jan;87(1):61-4. http://dx.doi.org/10.3357/AMHP.4421.2016 ; PubMed PMID: 26735235 , Jan-2016
NASA Technical Documents	Conkin J, Abercromby AFJ, Dervay JP, Feiveson AH, Gernhardt ML, Norcross J, Ploutz-Snyder R, Wessel JH 3rd. "Probabilistic assessment of treatment success for hypobaric decompression sickness." Houston, Tex. : NASA Johnson Space Center, 2014. NASA Technical Publication NASA/TP-2014-218561. https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20140012081.pdf ; accessed 3/14/17. , Nov-2014